

Data Analysis (Train: trainset_synth, Test: feb2026rb2r2)

The data analysis is based upon the predictions.csv files generated by 3 models viz. DNN, DeepNN and LSTM for voltage and power respectively.

Summary: Across all models, prediction error is consistently driven by load_current and current, showing strong positive correlation, especially for DNN and DeepNN. Feedforward models (DNN, DeepNN) exhibit high errors and struggle with nonlinear behavior and extreme load conditions particularly in power prediction. In contrast, LSTM significantly reduces errors and shows stable performance, indicating effective capture of temporal dynamics and system behavior. Overall, model performance improves from DNN to DeepNN to LSTM, with LSTM being the most reliable across both voltage and power tasks.

Test Data Distribution:

current	heater_power	load_current	eclipse_flag	payload_status
Mean: 3.8349	Mean: 215.1674	Mean: 9.8233	Mean: 0.7486	Mean: 0.2266
Std : 5.5261	Std : 87.3810	Std : 3.5540	Std : 0.4338	Std : 0.4187

Range:

current	heater_power	load_current	eclipse_flag	payload_status
-0.0531 → 87.3413	10.8000 → 446.8000	4.1735 → 88.8946	0.0000 → 1.0000	0.0000 → 1.0000

1. Voltage DNN:

Feature vs Error Correlation:

current	: 0.3772
heater_power	: -0.1002
load_current	: 0.7883
eclipse_flag	: -0.0228
payload_status	: 0.0384

Error by feature bins:

Feature	Average Error
current	21.7590
heater_power	17.6375
load_current	16.6674
eclipse_flag	17.0260
payload_status	17.0260

The model’s error shows a strong positive correlation with **load_current (0.78)**, indicating that prediction accuracy degrades as load increases. Moderate correlation with **current (0.37)** further suggests sensitivity to operating conditions. Error-by-bin analysis reveals the highest error in **high current regions (~21.7)**, confirming poor handling of extreme values. Overall, the model **struggles with nonlinear behavior** and **skewed data containing rare high-load spikes**.

2. Voltage DeepNN:

Feature vs Error Correlation:

current	: 0.3844
heater_power	: -0.0687
load_current	: 0.8219
eclipse_flag	: -0.0189
payload_status	: 0.0706

Error by feature bins:

Feature	Average Error
current	18.9955
heater_power	13.6817
load_current	13.8490
eclipse_flag	13.3008
payload_status	13.3008

The DeepNN model shows a strong positive correlation with **load_current (0.82)** and moderate correlation with **current (0.38)**, indicating error increases with higher load conditions. Compared to DNN, overall errors are reduced (~13–19), showing improved learning. However, the highest error still occurs in **high current regions (~19)**, indicating continued difficulty in handling extreme values. Thus, while DeepNN improves accuracy, it still struggles with **nonlinear behavior under high-load conditions**.

3. Voltage LSTM:

Feature vs Error Correlation:

current	: 0.1469
heater_power	: -0.2668
load_current	: 0.2291
eclipse_flag	: -0.0238
payload_status	: 0.0286

Error by feature bins:

Feature	Average Error
current	1.4725
heater_power	1.5660
load_current	1.5058
eclipse_flag	1.5386
payload_status	1.5386

The LSTM model shows weak positive correlation with **load_current (0.23)** and **current (0.15)**, indicating minimal error dependence on load conditions. Negative correlation with **heater_power (-0.27)** suggests slight stabilization in certain regimes. Error values are very low (~1.47–1.56) across all features, showing consistent performance. Overall, LSTM effectively captures temporal patterns and handles nonlinear behavior, significantly outperforming DNN and DeepNN.

4. Power DNN:

Feature vs Error Correlation:

current	: 0.3614
heater_power	: -0.0757
load_current	: 0.8107
eclipse_flag	: 0.0069
payload_status	: 0.0589

Error by feature bins:

Feature	Average Error
current	191.5049
heater_power	171.6738
load_current	168.7202
eclipse_flag	155.2454
payload_status	155.2454

The Power DNN model shows strong positive correlation with **load_current (0.81)** and moderate correlation with **current (0.36)**, indicating error increases significantly with load conditions. Extremely high bin errors (~155–191) reveal poor prediction accuracy, especially in high current regions. This indicates severe difficulty in capturing **power dynamics (V×I relationship)** under varying loads. Overall, the model struggles with **nonlinear interactions and error propagation in power prediction**.

5. Power DeepNN:

Feature vs Error Correlation:

current	: 0.3885
heater_power	: -0.1164
load_current	: 0.7677
eclipse_flag	: -0.0735
payload_status	: 0.0068

Error by feature bins:

Feature	Average Error
current	191.6556
heater_power	176.6691
load_current	136.7308
eclipse_flag	122.8142
payload_status	122.8142

The Power DeepNN model shows strong positive correlation with **load_current (0.77)** and moderate correlation with **current (0.39)**, indicating error increases with load intensity. While slightly improved over DNN in some regions (lower error for load_current), overall errors remain very high (~122–191). The highest error still occurs in **high current regions**, showing difficulty in modeling power under extreme conditions. Thus, DeepNN offers limited improvement and still struggles with **nonlinear power interactions and load-driven variability**.

6. Power LSTM:

Feature vs Error Correlation:

current	: 0.3681
heater_power	: -0.1753
load_current	: 0.7243
eclipse_flag	: -0.0254
payload_status	: 0.0862

Error by feature bins:

Feature	Average Error
current	29.7506
heater_power	20.2057
load_current	38.7372
eclipse_flag	18.4903
payload_status	18.4903

The Power LSTM model shows strong positive correlation with **load_current (0.72)** and moderate correlation with **current (0.37)**, indicating error still increases with load. However, errors are significantly reduced (~18–39) compared to DNN/DeepNN, showing much better performance. The highest error occurs in **load_current (~38.7)**, indicating some difficulty in extreme load regions. Overall, LSTM effectively captures temporal and nonlinear power behavior, though minor load-driven errors remain.